The Effects of a Visual Fidelity Criterion on the Encoding of Images

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Abstract—**Shannon's rate-distortion function provides a potentially useful lower bound against which to compare the rate-versus-distortion performance of practical encoding-transmission systems. However, this bound is not applicable unless one can arrive at a numerically-valued measure of distortion which is in reasonable correspondence with the subjective evaluation of the observer or interpreter. We have attempted to investigate this choice of distortion measure for monochrome still images. This investigation has considered a class of distortion measures for which it is possible to simulate the optimum (in a rate-distortion sense) encoding. Such simulation was performed at a fixed rate for various measures in the class and the results compared subjectively by observers. For several choices of transmission rate and original images, one distortion measure was fairly consistently rated as yielding the most satisfactory appearing encoded images.**

I. Introduction

SHANNON'S rate-distortion theory was originally developed to handle just such problems as the efficient encoding of images. Let $U(x,y)$ represent the intensity of an image scanned at the transmitter and $\tilde{U}(x,y)$ the intensity of the image displayed at the receiver. Let $d(\ ,\ )$ be a real-valued function of two images defined on a given raster. Consider $d(U,\tilde{U})$ to be the distortion that occurs when $U(x,y)$ is scanned at the transmitter and $\tilde{U}(x,y)$ is displayed at the receiver; the performance of the system is then measured by the average distortion

$$d^* = E\{d(U,\tilde{U})\}$$

that occurs in transmitting an ensemble of images. Shannon's rate-distortion function $R(d^*)$ is a lower bound on the transmission rate (in bits/unit area, bits/raster, or bits/picture element) required to achieve average distortion d^*; Shannon's coding theorem also states that one can design a code with rate only negligibly greater than $R(d^*)$ which achieves average distortion d^* [1], [2]. The function $R(d^*)$ thus exactly specifies the minimum achievable transmission rate R required to transmit an image with average distortion level d^*. The main value of this function is that it potentially gives one an *absolute* yardstick against which to compare the performance of a practical system and provides a useful basis for judging whether it is worthwhile to try to design a better system.

To date, this potential value has not been realized for image transmission. There are several reasons for this. One important one is that there do not currently exist tractable mathematical models for an image source (the Gaussian model is obviously a poor choice, even considering only first-order statistics). A second, and less important, reason is the difficulty of calculating the rate-distortion function for other than Gaussian sources and square-error distortion measures. However, we believe that the prime reason that rate-distortion theory is not currently applicable is that a distortion measure $d(\ ,\ )$ which is in accord with subjective evaluation of image quality is not known. That is, if $u(x,y)$ is an original image and $\tilde{u}_k(x,y)$, $k = 1,2,\cdots$, is any set of possible reproduced images, then $d(u,\tilde{u}_k)$ should rank the reproduced images in the same order as the photo interpreter or end user of the images. Until such a distortion measure can be found, calculation of rate-distortion functions will be mostly an academic exercise with little practical relevance to image transmission.

To obtain information concerning what properties such a function $d(u,\tilde{u})$ should have, at least two basic approaches are possible. In the first, one can make psychophysical measurements to find the level at which a given pseudorandom perturbation of the intensity pattern of a given meaningful image is just detectable by a human subject. Doing this for a wide variety of pseudorandom perturbation patterns and meaningful images gives useful information on the structure of $d(u,\tilde{u})$. In the second approach, which is the one we describe in this paper, one picks a class of distortion measures for which one can calculate $R(d^*)$ and simulate the optimum encoding. One can then compare different distortion measures in the class by simulating the encoding of a *fixed image* at a *fixed rate* in bits/picture element (hereafter abbreviated bits/pel) under different distortion measures and subjectively judging the quality of the encoded images. This method has the disadvantage that the study is restricted to only a certain class of distortion measures, but has the advantage of providing comparison of distortion measures directly in the context of encoding. If, for a variety of source images and all rates of interest, many subjects uniformly pick images encoded under the same distortion measure as appearing best, then clearly this measure is the most appropriate in the class to use for evaluating transmitted images.

In the second section we discuss the choice of the class of distortion measures. In the third, we describe the simulation. In the fourth section we present a sample of the encoded images and discuss the results of the judging.

II. The Class of Distortion Measures

The problem of finding a distortion measure that is in good accord with subjective evaluation is in its infancy.

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Some rudimentary facts are known: that the human observer is more sensitive to some spatial frequencies than others; that one is more sensitive to intensity errors in grey areas than in white; and, if $u(x,y)$ represents the intensity of a monochrome image, that

$$d(u,\tilde{u}) = \iint [u(x,y) - \tilde{u}(x,y)]^2 \, dx \, dy \tag{1}$$

is in very poor accord with subjective evaluation. Let us now give a quantitative description of what is known.

Consider an original image $u(x,y)$ and two possible reproductions $\tilde{u}'(x,y)$ and $\tilde{u}''(x,y)$. If $\tilde{u}''$ appears worse than $\tilde{u}'$, it may be difficult for observers to agree whether $\tilde{u}''$ is 3.5 times worse or 6 times worse. However, if the distinctions between u and $\tilde{u}'$ and between u and $\tilde{u}''$ are both just on the threshold of perception, it seems reasonable to say that $\tilde{u}'$ and $\tilde{u}''$ are equally bad; i.e., our distortion function $d(\ ,\ )$ should set

$$d(u,\tilde{u}') = d(u,\tilde{u}''). \tag{2}$$

Next let us describe some properties of the visual system obtained by psychophysical experiments on the visibility of gratings. Consider an intensity pattern of the form

$$u(x,y) = U_0 + \alpha p(x \cos\theta - y \sin\theta) \tag{3}$$

in which $p(\ )$ is a periodic function. Such an image has the appearance of a grating running at an angle of θ to the horizontal. For a variety of periodic functions $p(\ )$ and background levels U_0, experimenters [3]–[9] have measured the contrast sensitivity; that is, the value of the ratio U_0/α at which a subject can just detect the presence of the grating on the uniform background.

For horizontal sinusoidal gratings such that $p(x \cos\theta - y \sin\theta) = \cos[2\pi f x]$ measurements have been made for a number of values of f and it has been found that the contrast sensitivity ratio is of the form

$$\left.\frac{U_0}{\alpha}\right|_{\text{threshold}} \approx c(f/f_0)e^{-f/f_0}. \tag{4}$$

The value of f_0, the position of the peak of the curve, falls between 3 and 5 cycles per degree (per degree subtended in the subject's field of vision) and measurements have been made from frequencies of about 0.2 cycles/degree to 40 cycles/degree. For vertical gratings the contrast sensitivity is approximately the same. The contrast sensitivity decreases for gratings at an angle to the horizontal or vertical, being approximately 3-dB down at 45° to the horizontal [5]. The contrast sensitivity function varies with the background illumination level U_0, but for background illumination levels in the photopic range of vision the contrast sensitivity function varies only slightly with illumination level [4]. The photopic range of illumination is from a moderately illuminated room up to bright sunlight; in this range the cones of the retina are active rather than the rods and the visual system is capable of its highest resolution.

It is also a well documented fact that the visual system is not linear (see Marimont [10], for example). However, the results of a number of recent studies [11]–[14] seems to indicate that, after an initial nonlinear transformation, the remainder of the visual system may be considered linear over a moderate range of intensities.

Now let us consider a mathematical model that is consistent with these results. If $u(x,y)$ represents the intensity, set

$$w(x,y) = f[u(x,y)] \tag{5}$$

in which $f(\ )$ is a monotonic increasing convex $\cap$ function. We now wish to describe a linear spatially-invariant operation on $w(x,y)$. Just as a linear time-invariant operation on a time signal $w(t)$ can be described in terms of its impulse response $a(t)$ or transfer function $A(f)$, a linear spatially-(shift) invariant operation on a function of two variables may be described in terms of its point spread function $a(x,y)$ or transfer function $A(f_x,f_y)$. Let $v(x,y)$ be the result of operating on $w(x,y)$ with a linear spatially-invariant filter. The threshold measurements with sinusoidal gratings could then be modeled by a detection operation on $v(x,y)$.

This model is not intended to be phenomenological, i.e., to explain the physiological functioning of the human visual system. However, the action potential produced by the cones does have a monotonic convex relation to the incident light level. The linear operation has some correspondence to the lateral inhibition produced by neural interaction at the retinal level and above and to the limited resolution caused by the nonzero size and spacing of the photoreceptors. The former causes the visual system to act as a differentiator at low frequencies (the f/f_0 term in (4)) and the latter causes the rapid roll-off at high frequencies (the e^{-f/f_0} term in (4)).

To complete the model, we must specify some distortion measure on the transformed fields $v(x,y)$ and $\tilde{v}(x,y)$. The results of Campbell and Robson [3] and Sachs *et al.* [9] on more complicated periodic gratings suggest a mechanism which divides the transformed field into different frequency bands, measures the rms difference in each band, and or's the rms levels. Unfortunately, the only (to the author's knowledge) distortion measure for which it is possible to calculate the rate-distortion function and simulate the optimal encoding scheme is the integral square error, that is the energy in $v(x,y) - \tilde{v}(x,y)$. For this reason, we took $d(\cdot,\cdot)$ to be of the form

$$d(u,\tilde{u}) = \iint_{\text{raster}} [v(x,y) - \tilde{v}(x,y)]^2 \, dx \, dy. \tag{8}$$

The distortion measures that we considered are thus of the form (8), with $w(x,y) = f[u(x,y)]$, $\tilde{w}(x,y) = f[\tilde{u}(x,y)]$, and $v(x,y)$ and $\tilde{v}(x,y)$ the result of a linear spatially-invariant operation $A(f_x,f_y)$ on $w(x,y)$ and $\tilde{w}(x,y)$. Based on what is known physiologically and from psychophysical measurements, we restricted f to be monotonically increasing and convex $\cap$. Although it is known that the human visual system is not isotropic [5], most of the subject matter used in the experiments was isotropic. For example a moonscope can logically be viewed from any angle and in fact an observer might well rotate the image and view it from several angles. Moreover, for ease of implementation it was desired to minimize the number of parameters describing

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$A(f_x,f_y)$. For these reasons the linear operation was taken to be isotropic; i.e.,

$$A(f_x,f_y) = \tilde{A}(f_r), \qquad f_r = \sqrt{f_x^2 + f_y^2}. \tag{9}$$

To fit the data obtained from psychophysical measurements of sine-wave gratings, $\tilde{A}(f_r)$ would be almost of the form

$$\tilde{A}(f_r) = \frac{U_0}{\alpha}\bigg|_{\text{threshold}} = c(f_r/f_0)e^{-f_r/f_0}.$$

For this reason, we considered functions for $\tilde{A}(f_r)$ which fitted or nearly fitted the expression

$$\tilde{A}(f_r) = [c + (f_r/f_0)^{k_1}] \exp [-(f_r/f_0)^{k_2}]. \tag{10}$$

We thus investigated all distortion measures of the form (8) for $f(\ )$ monotonically increasing convex $\cap$ and different choices of the parameters f_0, c, k_1, and k_2.

III. Description of the Simulation

To calculate the rate-distortion function and be able to simulate the optimum encoding of images, we must specify the distortion measure *and* the probability distribution on the source output. In this case the source output is an image which is a function of two spatial variables x and y on some raster $0 \le x \le x_0$, $0 \le y \le y_0$. The (random) source output is thus a two-dimensional random field U_{xy} (a collection of random variables parametrized by two independent variables). To completely specify its probability distribution means that for an arbitrary n and an arbitrary choice of points $x_1, y_1; x_2, y_2; \cdots; x_n, y_n$, we must know the joint distribution of the n random variables

$$U_{x_k y_k}, \qquad k = 1,2,\cdots,n. \tag{11}$$

Next let us consider what is reasonable to assume known for an image source. For many images it may be reasonable to assume that the distribution is homogeneous, i.e., invariant to a shift in origin (corresponding to stationarity for a time process). Further it is reasonable to assume that by measurements one could estimate the mean

$$m = E\{U_{xy}\} \tag{12}$$

and correlation function

$$R_u(\tau_x,\tau_y) = E\{U_{x+\tau_x, y+\tau_y} U_{xy}\}. \tag{13}$$

It seems dubious that one could know much beyond this (it is easy to postulate good models for the one-dimensional processes formed by scanning images, but difficult to postulate two-dimensional models for the field). The only tractable model of a random field for which the mean and correlation function specify the joint distribution of the random variables in (11) is the Gaussian random field. Examination of the histogram from an image reveals that the probability density of U_{xy} is certainly not gaussian: typically there are two peaks, one in the black and the other in the white; other times the density may be broad and nearly uniform; in no case does one find a bell-shaped Gaussian density centered in the grey.

One is thus led to the painful conclusion that, at present, we are unable to specify the probability distribution of an image source. Is the situation hopeless? Not completely; we can rephrase the question to ask: what is the best that we can do with the limited knowledge at our disposal? Specifically let us denote by $\mathscr{A}$ the class of all possible probability distributions for the source. In the image transmission case $\mathscr{A}$ is the class of all probability distributions on a random field with given (measured) mean and correlation function. Suppose we ask for a *single* encoding–decoding system which will yield average distortion less than or equal to d^* for any source α in $\mathscr{A}$ and then ask what is the minimum rate that such a system can have. It is reasonable to define this rate as the rate-distortion function for the class $\mathscr{A}$ and to denote it by $R_{\mathscr{A}}(d^*)$. In distinction, the previously discussed rate-distortion function for a specific source α should be denoted $R_\alpha(d^*)$.

How can $R_{\mathscr{A}}(d^*)$ be determined? Clearly

$$R_{\mathscr{A}}(d) \ge \sup_{\alpha \in \mathscr{A}} R_\alpha(d^*) \tag{14}$$

since our single encoding system must be able to cope with the worst source in $\mathscr{A}$. Strict inequality is possible, since having to cope with *all* sources in $\mathscr{A}$ is a more difficult task than just coping with the worst source in $\mathscr{A}$. However, it turns out that for all physically meaningful classes of sources that [15], [16]

$$R_{\mathscr{A}}(d^*) = \sup_{\alpha \in \mathscr{A}} R_\alpha(d^*). \tag{15}$$

Thus calculation of the rate for a class of sources resolves to being able to calculate the rate-distortion function of the "worst" source in the class.

It is debatable as to what should be regarded as the rate of the class $\mathscr{A}$, since universal encoding methods exist [25], [26] that allow one to achieve the minimum distortion d consistent with the equation $R_\alpha(d) = R^*$ when the source α is to be transmitted at rate R^*. However, these encoding methods are nonconstructive in the following sense. The method of Ziv [25] involves collecting subblocks of source outputs and then collecting these into superblocks. Each time a superblock is collected, one must solve *from scratch*, the encoding problem for subblock length for the ensemble of subblocks composed of the superblock. The method of Gray *et al.* [26] involves essentially finding a set of sources dense in $\mathscr{A}$ and finding an efficient code of rate R^* for *each* of these sources. In the present context, neither method offers a possible method of solution for image encoding with only second moments known. By contrast, it is possible [20] for practical methods to come quite close to $R_{\mathscr{A}}(d)$. In this sense $R_{\mathscr{A}}(d)$ still seems like a more useful measure of the rate of the class $\mathscr{A}$.

If one assumes only that the second-order statistics (mean and correlation function or power density spectrum) are known, then for distortion measures of the class we are considering, the Gaussian source is the worst in the class [1]. Further, the optimum code for the Gaussian source which yields average distortion d^* for a Gaussian source is robust; i.e., it yields average distortion less than or equal to d^* for any source in the class [17]. We thus calculate the rate-distortion function for our class of source distributions as

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the rate-distortion function of the Gaussian source and simulate the optimum encoding for the Gaussian source.

At this point we should comment on why we simulate the optimum encoding. First, the number of different possible practical encoding schemes is unbounded; we could consider transform encoding for a large number of transforms and for each consider different truncation and quantization rules; we could consider DPCM and/or DPCM combined with one-dimensional transform encoding; we could consider a multitude of sampling and quantization schemes followed by error-free encoding. An exhaustive study is truly not possible. Moreover, since our goal is to study the role of the distortion measure in *efficient* encoding, the choice of simulating the *optimum* encoding is uniquely appropriate for our purposes. Thus our study was confined to simulating only the optimum encoding and did not consider simulating more practical methods (even simulating only the optimum encoding required producing over 300 images to cover the situations of interest). It must be conceded that the Gaussian distribution of encoding errors that result from simulating the optimum encoding is different from the distribution of errors resulting from practical schemes (say nearly uniform for quantization). However, in any transform method of encoding, the distribution of errors in the spatial domain will be nearly Gaussian (they will be sums of independent errors in the transform coefficients) and the difference in appearance due to different distributions of errors should be unimportant.

For a raster large compared to the "correlation distance" of $V(x,y)$, the rate-distortion function is given parametrically by the pair of equations [18]

$$R(\mu) = 1/2 \iint_{S_v(f_x,f_y)>\mu} \log_2 [S_v(f_x,f_y)/\mu]\, df_x\, df_y \quad (16)$$

$$d^*(\mu) = \int_{-\infty}^{\infty}\int \min [S_v(f_x,f_y),\mu]\, df_x\, df_y \quad (17)$$

in which $S_v(f_x,f_y)$ is the power spectral density of $V(x,y)$ (the two-dimensional Fourier transform of $R_v(\tau_x,\tau_x)$). For an isotropic (rotation-invariant) random field, $R_v(\tau_x,\tau_y)$ is a function only of $\tau_r = \sqrt{\tau_x^2 + \tau_y^2}$ and $S_v(f_x,f_y)$ is a function only of $f_r = \sqrt{f_x^2 + f_y^2}$. Defining $\tilde{S}(f_r)$ by

$$S_v(f_x,f_y) = \tilde{S}(f_r), \qquad f_r = \sqrt{f_x^2 + f_y^2} \quad (18)$$

(16) and (17) then become

$$R(\mu) = \pi \int_{\tilde{S}_v(f_r)>\mu} \log_2 [\tilde{S}_v(f_r)/\mu] f_r\, df_r \quad (16')$$

$$d^*(\mu) = 2\pi \int_0^{\infty} \min [\tilde{S}_v(f_r),\mu] f_r\, df_r. \quad (17')$$

Again for raster size large compared to the correlation distance, the optimum code operates as follows [19], [20]. The field $V(x,y)$ is expanded on its raster in a Fourier series expansion

$$V(x,y) = \sum_{k,j} V_{kj} \exp [i2\pi(kx/x_0 + jy/y_0)], \quad 0 \le x \le x_0 \quad 0 \le y \le y_0. \quad (19)$$

Those coefficients $V_{k,j}$ for which

$$E\{|V_{k,j}|^2\} \triangleq \lambda_{kj} < \mu \quad (20)$$

are not transmitted. The remaining coefficients are transmitted in such a fashion that the received coefficients $\tilde{V}_{k,j}$ have a distribution such that the complex random variables $V_{k,j} - \tilde{V}_{k,j}$ are Gaussian with zero mean and variance μ, and independent of $\tilde{V}_{k,j}$. This can be simulated [19] by setting

$$\begin{Bmatrix}\text{Re}\\ \text{Im}\end{Bmatrix} \tilde{V}_{k,j} = \frac{\lambda_{kj} - \mu}{\lambda_{kj}} \left[\begin{Bmatrix}\text{Re}\\ \text{Im}\end{Bmatrix} V_{k,j} + \begin{Bmatrix}\text{Re}\\ \text{Im}\end{Bmatrix} N_{k,j} \right] \quad (21)$$

in which $N_{k,j}$ are zero-mean complex-valued random variables which are independent of $V_{k,j}$, whose real and imaginary parts are uncorrelated, and each of which has variance $\mu\lambda_{kj}/2(\lambda_{kj} - \mu)$.

The simulated decoded image is then constructed as a Fourier series expansion in the coefficients $\tilde{V}_{kj}$

$$\tilde{V}(x,y) = \sum_{\substack{k,j \\ \lambda_{k,j}>\mu}} \tilde{V}_{kj} \exp [i2\pi(kx/x_0 + jy/y_0)], \quad 0 \le x \le x_0 \quad 0 \le y \le y_0. \quad (22)$$

From this field the decoded image is then constructed by performing the inverse of the linear operation $A(f_x,f_y)$ and the inverse of the nonlinear operation $f[\ ]$ to obtain $\tilde{U}(x,y)$.

Our discussion to this point has discussed the image as being a function of the continuous spatial parameters x and y. However, the simulation program must operate on images defined on a discrete raster of 512 × 512 picture elements (pels). The two viewpoints are not in contradiction, since the simulated images were viewed at a distance at which 65 pels subtended 1° of vision. At this distance the observer cannot distinguish the discrete raster on which the image is defined from a continuous one. To perform the simulation expressed by (22), the program executes the following computations on a 512 by 512 raster for a given image and transmission rate and a particular choice of $f(u)$ and $\tilde{A}(f_r)$.

1) Calculate $w(x,y) = f[u(x,y)]$ and estimate the power spectral density of $W(x,y)$ from $w(x,y)$ using a smoothed periodogram technique. It was assumed that the field was isotropic so $S_w(f_x,f_y) = \tilde{S}_w(f_r)$, with $f_r = \sqrt{f_x^2 + f_y^2}$. This ignores, among other things, edge effects which make the image nonisotropic. In estimating the spectrum, we attempted to minimize the effects of the finite raster size. Moreover, the displayed image subtended 6°, while an observer's detailed vision subtends roughly 1°, so that edge effects were not crucial in judging.

2) Compute $\tilde{S}_v(f_r) = |\tilde{A}(f_r)|^2 \tilde{S}_w(f_r)$ and from this find the value of the parameter μ in (16) corresponding to the given rate.

3) Calculate $W(f_x,f_y)$, the fast Fourier transform (FFT) of the 512 × 512 array $w(x,y)$ and then calculate $V(f_x,f_y) = \tilde{A}(f_r)W(f_x,f_y)$ for $f_x = k/x_0$, and $f_y = j/y_0$, where $x_0 = y_0$ is the width of the square raster. W and V are 256 × 512 complex arrays, since they are conjugate symmetric.

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4) Those coefficients $V_{kj} = V(k/x_0, j/x_0)$ for which

$$\lambda_{kj} = E\{|V_{k,j}|^2\} > \mu$$

have independent samples of Gaussian noise added to them and are scaled according to (21) to yield coefficients $\tilde{V}_{k,j}$. The remaining coefficients are set equal to zero.

5) The Fourier transform of $\tilde{w}(x,y)$ is computed by filtering the $\tilde{V}_{kj}$ by the inverse of $\tilde{A}(f_r)$

$$\tilde{W}_{kj} = \tilde{V}_{kj}/\tilde{A}[(k^2 + j^2)/x_0^2].$$

6) The field $\tilde{w}(x,y)$ is obtained by taking the inverse FFT of the array $\tilde{W}_{kj}$ and the image $\tilde{u}(x,y) = f^{-1}[\tilde{w}(x,y)]$ is generated and recorded photographically using a high-resolution CRT display.

In the next section, we present results of the simulation and discuss the judging of the simulated images encoded according to different choices for $f(\ )$ and $\tilde{A}(f_r)$. For a more detailed description of the simulation routines, the reader is referred to Mannos [21].

IV. Results of the Simulation

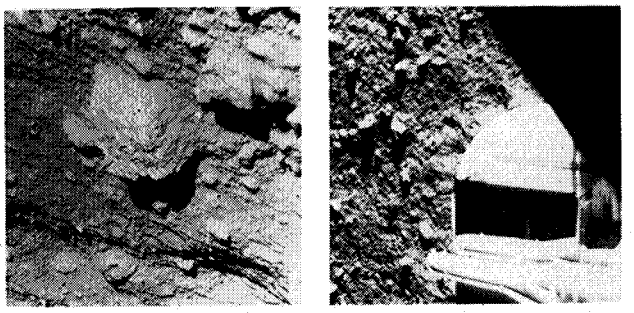
In this section we describe the results of the simulation. We start by briefly describing the experimental technique used to select the parameters in the distortion measure.

First an original test image, a fixed transmission rate R, and a distortion measure of the form described in Section II are chosen. Then, following the methods described in the last section, the rate-distortion test channel is simulated on the digital computer. This produces an image distorted in exactly the same manner as if the optimum encoder had been used to transmit the original image over a channel of capacity R. If this experiment is repeated for different choices of the parameters describing the distortion measure, we obtain a collection of pictures, each one transmitted over the channel of capacity R, but appearing different due to the fact that each encoding scheme emphasizes those portions of the images which the visual model implicit in the distortion measure considers important. Then a group of observers is asked to rank the pictures subjectively in terms of their closeness to the original image. The results of this judging are tabulated to determine if any particular model is consistently selected as best by the observers. Next, the experiment is repeated for a different rate R and/or a new original image. These results are also tabulated and then compared with those of the previous experiments. If any one set of model parameters is consistently better in appearance, independent of the rate or picture selected, then these particular parameters are to be chosen for the visual model. This procedure resulted in the generation of approximately 360 different pictures.

Before discussing the results of the experiments it would be useful to describe in more detail the judging procedure and images used.

A. Experimental Procedure

Fig. 1 shows the set of three original pictures used in the experiment. The "rock" picture has a large amount of very detailed structure over the whole picture area, and an intensity level predominantly in the middle greys. The



(a) (b)



(c)

Fig. 1. Pictures used for experiments.

"footpad" picture contains less detail than the "rock" and large areas of both deep black and bright white. Its structure is considerably less uniform than the rock and contains the sharp straight lines associated with manmade objects. The face picture "Ronnie" contains the least amount of detail, but would be the most familiar to an observer. It was felt that these pictures allowed for sufficient variety to test the sensitivity of the model parameters to image content.

In order to keep unwanted photographic variations from interfering in the image quality, the exposure and development of the negatives were checked very carefully. Only fresh chemicals and film were used, and the temperatures of the photographic solutions were kept to within $\frac{1}{2}$°F by a regulator valve. In addition, a step grey scale was scanned into each negative and checked with a densitometer. The photographic images were enlarged onto a 4 in. × 5 in. sheet of Agfa BW-111 paper and dried to a semigloss finish. Due to borders around the photograph, the actual image size was 3.75 × 3.75 in. Reflection measurements from the deepest black and lightest white which the paper could produce indicated a maximum contrast ratio of 32 to 1.

At a viewing distance of 36 in. the images subtend a visual angle of 6°. All the spatial frequencies then may be

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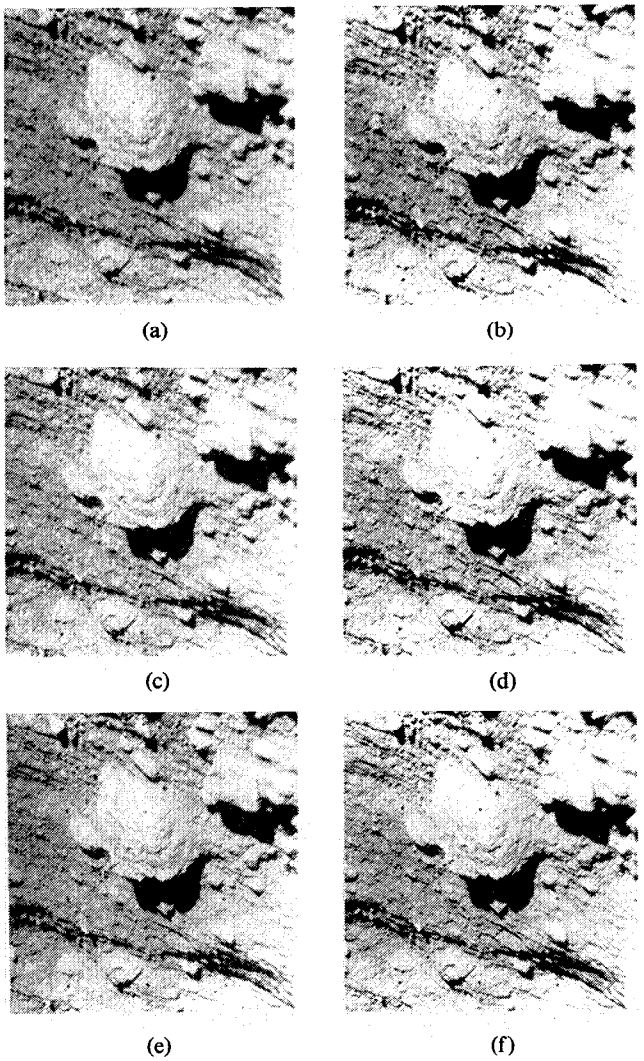


Fig. 2. Images obtained from simulated optimum encoding. Pictures (a)–(c) are with $f(u) = u$, $A(f_r) \equiv 1$; (d)–(f) are with optimum choices determined for $f(\,)$ and $A(f_r)$. Rates are (a) and (d)—½ bit/pel; (b) and (e)—1 bit/pel; (c) and (f)—2 bits/pel.

expressed in standardized units of cycles/degree. This viewing distance also creates just under 1 diopter of accommodation in the observer's visual system.

Illumination of the photographic images was provided by four 100-W incandescent flood lamps. The photographs were mounted on 4 ft × 6 ft white matte board to provide a large uniform surround. Average luminance was about 200 ml giving rise to an approximately 3.0 mm pupil size.

Nine subjects were used in the judging. These were mainly graduate students in the picture processing area and two subjects with no particular interest in picture processing: a speech therapist and an electronic technician.

Judging procedure began with the subjects allowing their eyes to adapt to the viewing conditions. Then from 4 to 8 pictures would be placed on the large white surround along with an original copy of the picture. The subjects were then asked to rank the pictures in their overall closeness in appearance to the original. A technique known as a "bubble sort" was used. The subjects would compare two pictures, say A and B, and decide which was most nearly like the original. Then another picture C could be selected and compared with A; if C was then considered better than A, the pictures would be placed in the order CAB. If A was better than C, the order would be ACB and a C versus B comparison would be made to determine the final order. This procedure would continue until all the pictures had been judged and placed in order. Thus the best pictures would "bubble" to the top of the ranking and the worst to the bottom.[1] Each pairwise test required a forced choice, i.e., no ties were allowed. As a further test the subjects were asked to evaluate each picture on the following scale of from 1 to 7:

7 best in group;
6 well above average for this group;
5 slightly above average for this group;
4 average for group;
3 slightly below average for this group;
2 well below average for this group;
1 worst in group.

In this case ties are allowed. Note that the scale is not absolute, but relates only those pictures within a group. This is necessary because pictures produced at different rates vary considerably. See Fig. 2 (a)–(c) for example, where the rock picture is shown for several different transmission rates expressed in bits/pel (picture element). This rapid change in quality with rate is typical of virtually all the models, and would result in placing most pictures produced at a given rate in a single category on an absolute scale, a clearly undesirable effect.

The addition of a scaling technique to the "bubble sort" ranking was intended to serve two purposes. First it was a check on the ranking since both techniques in fact form a rank order and should agree. And second, it performed a sensitivity test since ties were allowed. For example, the forced-choice decisions used with the bubble sort might in fact cause one picture to be ranked consistently ahead of another, even when the differences in picture quality are very small. However, the scale values for the two pictures would be very close, reflecting the fact that the choice between the images is not nearly so overwhelming as the ranking differences might imply.

In the selection of visual models for use in the tests it is desirable to have a wide variety of different types. However, each model would have to be tested for all the original pictures and for a variety of different transmission rates. This can very quickly produce an unmanageable number of pictures, particularly if it is realized that each picture requires an hour of computer time and then must be judged against all the other models by nine different observers.

To minimize this problem the first set of pictures were produced with widely varying properties in order to gauge

[1] Considerable effort has gone into devising methods of scaling and ranking for use in subjective tests [22], [23]. However, almost all of these methods make assumptions, some of which seem rather dubious, on the mathematical models underlying the judging process. The "bubble sort" technique seemed to be a good compromise and worked consistently with the subjects used in the judging process.

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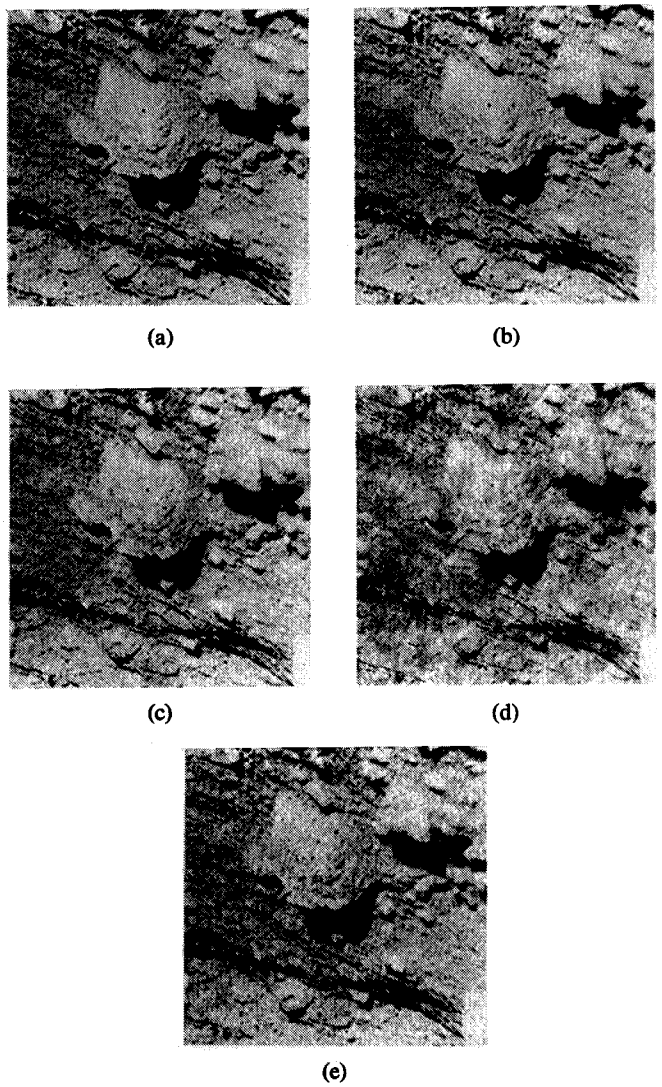


Fig. 3. Examples of simulated encoded pictures for different peak frequencies of $A(f_r)$ in cycles/degree. Rate = $\frac{1}{2}$ bit/pel. (a) 2. (b) 4. (c) 8. (d) 16. (e) $A(f_r) \equiv 1$.

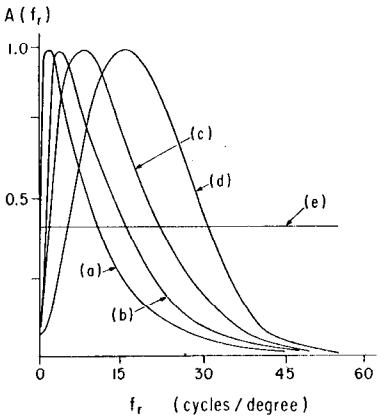


Fig. 4. Plots of $A(f_r)$ curves used to generate simulated images of Fig. 3.

TABLE I
RESULTS OF JUDGING TESTS FOR VARIATION IN PEAK FREQUENCY OF $A(f_r)$

Peak Freq. cycles/degree	Score in the Ranking test	Variance of Rank	Scale Value Score	Variance Scale Val.
.5 bits/pel rock picture				
0	NJ			
1	NJ			
2	7.00 (8th)	1.55	2.44	0.69
4	5.88 (6th)	1.65	2.77	0.61
6	4.55 (4th)	1.35	3.77	0.83
8	1.66 (1st)	1.11	6.33	0.66
10	2.11 (2nd)	0.54	5.88	0.32
12	3.88 (3rd)	4.32	4.44	3.13
16	6.00 (7th)	5.55	2.66	3.99
25	NJ			
45	NJ			
uniform	4.88 (5th)	1.20	3.55	0.69
.5 bits/pel footpad picture (partial)				
4	4.00 (4th)	1.75	4.37	0.98
6	3.00 (2nd)	1.50	5.12	1.10
10	3.62 (3rd)	1.48	4.62	0.48
16	6.50 (7th)	1.75	2.50	1.00
.5 bits/pel Ronnie picture				
4	4.00 (4th)	0.00	2.50	0.50
6	2.25 (2nd)	0.66	4.25	1.68
8	1.12 (1st)	0.33	5.62	0.98
10	2.62 (3rd)	0.45	4.00	0.25
1.0 bits/pel rock picture (partial)				
4	5.62 (6th)	1.48	3.12	0.85
6	3.87 (3rd)	1.60	4.12	1.85
10	1.75 (1st)	0.43	6.25	0.43
16	6.25 (8th)	5.68	2.62	3.73

NJ: Not judged because image quality was considerably below average for this group.

(partial): Indicates that the images in this group are only a partial set of images taken from a larger judged group.

the relative importance of the different kinds of parameter variations. On the basis of these results, it was decided to divide the $A(f_r)$ curve up into three portions for the low, middle, and high frequencies. In particular, changing the position of the central peak of the curve by changing f_0 in (10) produced a very dramatic effect on image appearance, while changing the zero-frequency intercept by changing c and changing the high-frequency tail by changing k_2 had a secondary effect. Additionally, the linear and log functions were initially tested for the transformation f, with the log appearing better in every case.

Under these circumstances the first experiment was performed for a logarithmic choice of f, and with a variation of the middle-frequency peak.

B. Middle Frequencies: Variation of Central Peak of $A(f_r)$

Fig. 3 illustrates the effect of moving the position of the central peak of $A(f_r)$ for the $\frac{1}{2}$ bit/pel rock picture.[2] All comparison pictures shown in this paper are $\frac{1}{2}$ bit/pel, it being anticipated that at higher transmission rates the differences between images would be washed out in the

[2] The difference between the different pictures is considerably reduced due to the photoreproduction process used in printing. Prints of Figs. 1, 3, 5, and 6 may be obtained for $1.00 each by writing Miss Doris Simpson, E.R.L., Cory Hall, University of California, Berkeley, Calif. 94720.

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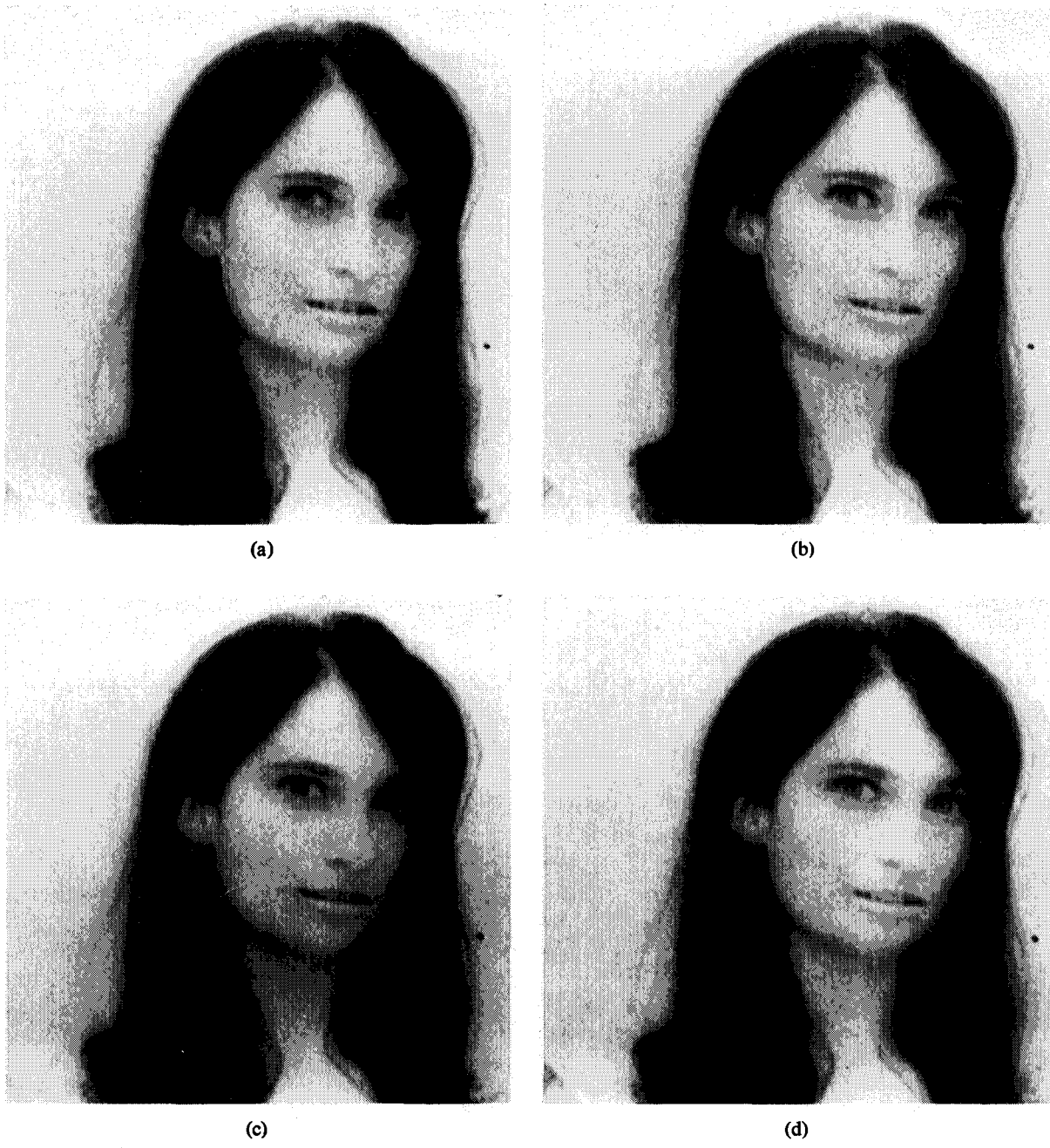


Fig. 5. Examples of simulated encoded pictures for different peak frequencies of $A(f_r)$ in cycles/degree. Rate = $\frac{1}{2}$ bit/pel (a) 4. (b) 6. (c) 8. (d) 10.

printing process. The $A(f_r)$ curves used in the generation of these images are shown in Fig. 4. The nonlinear portion of the model was chosen as logarithmic and the zero-frequency intercept $A(0)$ was made equal to 0.025. The high-frequency tail was tapered off to approximately zero at 60 cycles/degree. As might be expected, positioning the peak at the low frequencies below 4 cycles/degree gives rise to a distinctly blurred image. The subjects referred to a general fuzziness and lack of fine detail. Also evident is a distinct medium-high frequency noise which tends to obscure detail and give the picture an overall "dirty" appearance. At the higher frequencies of 12 cycles/degree and greater, a different phenomenon appears. Here the edges remain sharp and detailed, but the image takes on an overall "blotchy" appearance. These blotches are due to the large amount of low frequency noise introduced by the simulated encoding and become more pronounced as the peak of $A(f_r)$ is placed at higher frequencies. In the range of 4–12 cycles/degree these two effects are most nearly balanced. All the subjects selected a picture from this range as appearing most like the original.

The results of the judging are shown in Table I. The rock picture was tested at $\frac{1}{2}$, 1, 2, and 3.5 bits/pel. In all but the 1 bit/pel case the peak was selected at 8 cycles/degree with 10 cycles/degree finishing a close second. In the 1 bit/pel case 10 cycles/degree was chosen first and 8 cycles/degree second. However, the difference in scale values was small, i.e., 6.25 versus 6.00. At 3.5 bits/pel, most subjects felt that

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they could not choose accurately between the pictures, and scale scores of mostly fours were not uncommon. If forced to make a selection, most subjects picked the 10 or 12 cycle/degree images. The uniform frequency weighting (mean-square case) generally finished considerably behind the middle frequency peak but ahead of the high- and low-frequency peaks. It typically received a scale value of 3 to 3.5. However, its position in the rankings got worse with increasing bit rate. By 3.5 bits/pel it was a distinct last-place choice for all the subjects.

The footpad picture follows nearly much the same pattern as the rock picture. For the different transmission rates of $\frac{1}{2}$, 1, and 2 bits/pel, the 8 cycles/degree position for the peak was again determined to produce an image most nearly like the original. Second choice, however, was 6 cycles/degree, which is in contradistinction to the rock picture. This could very well be a reflection of the fact that the footpad contains less detail than the rock. Another interesting difference between the two images is the ranking of the footpad mean-square (uniform weighting) case consistently last. The subjects professed somewhat more difficulty in judging the footpad picture due to the three distinct areas—deep black shadow, a light white footpad, and high detail grey moon rocks—all of which must be taken into account. Again the 3.5 bits/pel rate produced pictures of nearly equal quality, with first place and fifth place being separated by scale values of just 4.87 to 4.25. The ranking jumped around from 16 cycles/degree to 6 cycles/degree, all with nearly equal scale values. It was felt that at 3.5 bits/pel picture quality was sufficiently good for all models as to make the judging of dubious value. Thus the remaining experiments were carried out for 2 bits/pel or lower transmission rates.

Fig. 5 shows the effect of varying f_0 for the image of Ronnie. The picture of Ronnie contains less detail than either of the other two images, but observers were able to make very rapid decisions between the choices, probably due to their familiarity with judging faces from everyday experience. The 8 cycles/degree peak position was overwhelmingly chosen first, followed by the 6 cycles/degree position. The low-frequency peak position was considered particularly objectionable due to the blurring of important facial features and the additional "dirty" noise effect on skin texture. For this picture the subjects had difficulty in making judgments at the 2 bit/pel rate, as image quality at this rate was virtually the same for all models; hence the judging was carried out only for the $\frac{1}{2}$ and 1 bit/pel cases.

The general conclusion that may be drawn from this set of experiments is that for a variety of subject matter and transmission rates, the $A(f_r)$ function should be chosen with a peak in the 8 cycles/degree range.

C. Low Frequencies: Selection of the Zero-Frequency Intercept

Variations in the low frequency portion of the model gave rise to less pronounced effects in the images. The first experiment was carried out for the rock picture by choosing the peak of $A(f_r)$ at 8 cycles/degree. The high-frequency curve was tapered off exponentially to approximately zero at 60 cycles/degree. Then if the peak height is normalized to 1, the various values 0, 0.025, 0.05, 0.1, 0.25, 0.33, and 1 were used for $A(0)$. As in the previous case we have two conflicting phenomena. First, as the intercept value is raised, more ephasis is placed on the low frequencies at the expense of the high frequencies and the image becomes more blurred and noisy. However, if the intercept is decreased below a certain point, then an uneven low-frequency shading occurs. We have not included examples of the resulting images because it was felt that the subtle differences between different values of $A(0)$ would not show up after the printing process. The results of the judging are shown in Table II; as can be seen from the table, the best compromise occurred for an intercept value of 0.05. Second choice was the 0.1 value, which usually ranked about 1 point lower in the scale value. The 0 and 1 values finished last and next to last, respectively.

TABLE II
RESULTS OF JUDGING TESTS FOR VARIATION IN $A(0)$

.5 bit/pel rock picture

A(0)	Score in the Ranking test	Variance of Rank	Scale Value Score	Variance Scale Val.
0	6.12 (7th)	0.60	2.25	0.68
.025	4.37 (4th)	0.85	3.75	0.18
.05	1.12 (1st)	0.10	5.75	0.68
.10	2.37 (2nd)	0.23	4.62	0.48
.20	3.00 (3rd)	1.50	4.50	0.50
.33	4.62 (5th)	1.73	3.37	0.48
1.0	6.00 (6th)	1.00	2.87	0.60

The variation in the intercept value was also done for $A(f_r)$ curves with central peaks at 6 and 10 cycles/degree, and the same effects were observed. However, the difference between intercept values was much less than the difference due to changing the position of the central peak, and at a given intercept value the 8 cycles/degree picture was always better than either the 6 or 10 cycle/degree ones.

In previous experiments all the curves were approximately linear (straight line) for the low frequencies between 0 and the central peak. On the 8 cycle/degree picture the low-frequency portion was also tried as an approximately f_r^2 function leading to more "blotchy" results. This particular experiment was not pursued any further, and an intercept of 0.05 and a linear function were chosen for the low-frequency portion of $A(f_r)$.

Similar results were obtained with both the Ronnie and footpad pictures.

D. High Frequencies: Variation in Exponential Decay

By the high frequencies we mean those values from the position of the central peak to approximately 60 cycles/degree. The rate of decay of the exponentially decreasing portion of the model is varied by varying the parameter k_2 in (10). For values of k_2 less than 1.5, the effect of the decay rate was small. Decays faster than 1.5 were not used in the judging since they considerably increased the overall

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TABLE III
RESULTS OF JUDGING TESTS FOR CHANGE IN POWER OF ARGUMENT OF EXPONENTIAL

.5 bit/pel rock picture

k_2	Score in the Ranking test	Variance of Rank	Scale Value Score	Variance Scale Val.
0.6	2.25 (3rd)	0.68	4.50	0.75
0.9	2.12 (2nd)	0.60	5.00	0.50
1.1	1.75 (1st)	0.93	5.50	1.25
1.5	4.25 (4th)	0.43	3.37	0.23

fuzziness of the image without adding any noticeable improvement in other aspects of the picture.

The central peak was fixed at 8 cycles/degree and $A(0)$ was held constant at a value equal to 0.05. Then the decay rate was progressively changed, using the values $k_2 =$ 0.6, 0.9, 1.1, and 1.5. This experiment also did not cause a dramatic difference in image quality. For the $\frac{1}{2}$ and 1 bit/pel rates the subjects could find discernible differences in the image quality, choosing a value of $k_2 = 1.1$ followed by the value 0.9. At 2 bits/pel the same order held, but by a very narrow scale value difference between the first two models (for example, 5.12 to 4.87). The one exception to this order was for the $\frac{1}{2}$ bit/pel footpad picture where $k_2 = 0.9$ was chosen first and $k_2 = 1.1$ second by the judging. However, the scale values for these two values of k_2 were fairly close together at 5.37 and 5.00, respectively.

While the effect of too rapid a decay as in (*d*) is a blurred picture, the slow decay of (*a*) gives rise to the usual problem of too much high-frequency emphasis, an uneven shading or "blotchy" effect. These effects were similar for all three test images. The results of the judging are shown in Table III.

In the next section we explore the variation in image characteristics for several choices of the nonlinear transformation f.

E. Nonlinear Portion of the Model

All the previous experiments were carried out assuming that the nonlinear portion of the model was logarithmic, or, more specifically, $\ln(1 + u)$ to avoid the singularity of $\ln(u)$ at $u = 0$. A few of the experiments were also done with the transformation $f(u) = u$ (linear). However, in these cases, the log transform appeared to give considerably better results.

In the experiments described in the following, the linear portion of the model was chosen to be the $A(f_r)$ judged best in the previous sections. This was a central peak position at 8 cycles/degree, a zero-frequency asymptote at 0.05 and a value of k_2 of 1.1.

In addition to the logarithmic and linear functions, two others were used in the test: $f(u) = u^{0.33}$ and $f(u) = u^{0.7}$. All the functions are strictly convex $\cap$ with the exception of the $f(u) = u$ transformation. The effect of this type of convexity is to expand and emphasize the dark areas at the expense of the light areas. This is a logical step since the eye is known to be more sensitive to small variations in dark surrounds than in light ones. Over the 0 to 63 intensity range of the test pictures, the $u^{0.33}$ function gives a more moderate low intensity expansion than does the logarithm. The $u^{0.5}$ and $\log u$ functions give virtually identical expansion over this range of intensity values, and hence the 0.5 exponent was not used in the tests. A $u^{0.7}$ function expands and emphasizes the dark areas in an even more pronounced manner than the logarithm and thus was added as a further test function.

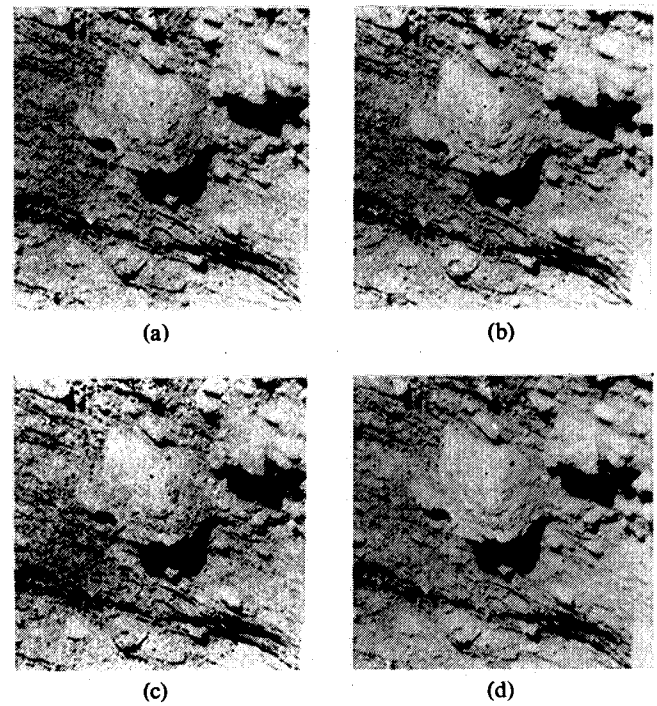


Fig. 6. Examples of simulated encoded pictures for various choices of $f(\,)$. Rate = $\frac{1}{2}$ bit/pel. (a) $f(u) = \log u$. (b) $f(u) = u^{0.33}$. (c) $f(u) = u^{0.7}$. (d) $f(u) = u$.

TABLE IV
RESULTS OF JUDGING TESTS FOR CHANGES IN NONLINEAR FUNCTION $f(u)$

$f(u)$	Score in the Ranking Test	Variance of Rank	Scale Value Score	Variance Scale Value
0.5 bits/pel rock picture:				
$\log u$	2.33 (2nd)	0.66	4.33	1.33
u	2.88 (3rd)	1.20	3.44	1.13
$u^{0.33}$	1.33 (1st)	0.22	5.44	0.91
$u^{0.7}$	3.44 (4th)	0.68	3.11	0.76
1.0 bits/pel rock picture:				
$\log u$	2.22 (2nd)	0.17	4.77	0.39
u	3.11 (3rd)	0.98	3.66	1.33
$u^{0.33}$	1.11 (1st)	0.09	5.77	0.39
$u^{0.7}$	3.55 (4th)	0.24	3.44	0.24

Fig. 6 shows simulated encoded images demonstrating the effects of varying the nonlinear portion of the model for the $\frac{1}{2}$ bit/pel rock picture. The results of the judging for three transmission rates ($\frac{1}{2}$, 1, 2 bits/pel) are shown in Table IV. The $u^{0.33}$ and $\ln(1 + u)$ functions were chosen first and second, respectively. The preference for $u^{0.33}$ at the $\frac{1}{2}$ and 1 bit/pel rates was substantial (typically scale values of 5.8 to 4.8), but less pronounced at 2 bits/pel (5.22 to 5).

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All the convex ∩ functions have the effect of diminishing noise in the dark shadow areas below the noise level present when the linear model was used. They also appeared to enhance the contrast of the image vis-a-vis the linear case, which appeared "washed out" or flat. However, the more convex the function became, the more the noise or "dirty" pattern began to show up in the light grey rock areas. The $u^{0.33}$ function was selected by the observers as being the best balance between these two effects, producing a picture most like the original.

With the footpad pictures the $u^{0.33}$ model was also judged best; however, the second choice was nearly a tie between the remaining three models. The subjects indicated that this was one of the most difficult pictures to judge because models strongly affected the large deep black and light white areas differently. As might be expected, those models which did well in the black areas did poorly in the white, and vice versa. While the $u^{0.33}$ function seemed to give the best balance, second choice seemed to be largely a matter of personal taste as to the relative importance of the various picture areas.

In the picture of Ronnie, the first and second choices were $u^{0.33}$ and logarithm, similar to the rock picture. The $u^{0.7}$ model was particularly disliked for a "measles" effect on the facial areas. It appears that on facial pictures the "dirty" effect of high-frequency noise is quite objectionable.

V. Conclusions and Possible Extensions

The results of the judging lead to a transfer function of approximately the form

$$A(f_r) \approx 2.6[0.0192 + 0.114f_r] \exp[-(0.114f_r)^{1.1}]. \quad (23)$$

This function has a peak of value 1.0 at $f_r = 8.0$ cycles/degree and a zero-frequency intercept of 0.5. The nonlinear function f is of the form $f(u) = u^{0.33}$. The final functions $A(f_r)$ and $f(u)$ resulting from the judging are shown in Fig. 7.

If the previous expressions for $A(f_r)$ and f are compared with the psychophysical experiments discussed in Section II, we notice considerable similarity. The bandpass form of $A(f_r)$, with a central peak at 8 cycles/degree and a rapid decrease on either side of this peak, is typical of the contrast sensitivity functions obtained from psychophysical experiments. The graph in Fig. 8 shows a plot of $A(f_r)$ superimposed on the contrast sensitivity functions of [3]–[8]. Both the low- and high-frequency portions of $A(f_r)$ seem to decay in a manner quite similar to these contrast sensitivity functions. The central peak position lies a little to the high-frequency side when compared to most of the contrast sensitivity functions. However, it is still less than that determined by some experimenters. All in all, there were no particular surprises in the linear portion of the model. However, the nonlinear portion did differ from what the authors expected: the choice of the $u^{0.33}$ function ahead of the logarithm. Although Stevens [24] had argued quite strongly for a power law with an exponent in this range, most other work has tended to support a logarithmic transformation. It might be noted that while the logarithm

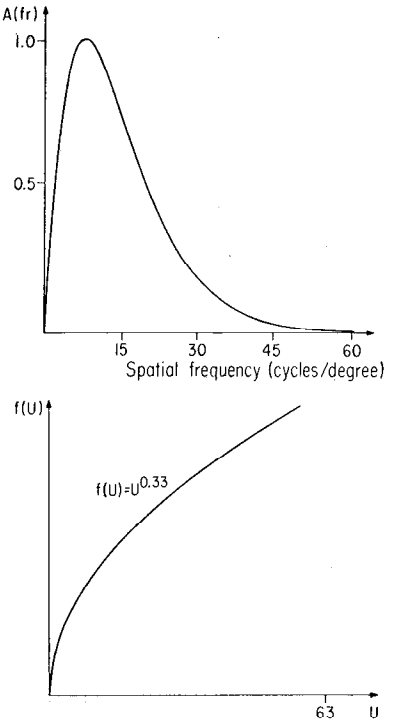


Fig. 7. $A(f_r)$ and $f(u)$ curves which produced simulated images judged to be best.

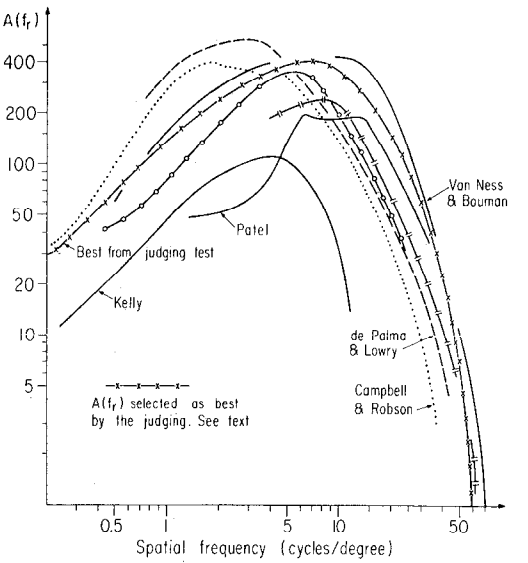


Fig. 8. $A(f_r)$ compared with contrast sensitivity measurements of sinusoidal gratings.

was selected as second choice, it was a distinct second in terms of the scale values.

It would be interesting at this point to compare, for the same rate, pictures transmitted using the best model and those transmitted using an unweighted (or uniformly weighted) linear model. This latter case corresponds to $A(f_r) = 1$ and $f(u) = u$ and is the standard method by which pictures are now transmitted. As may be seen from Fig. 2(a)–(f), the images produced by using the best model in the transmission scheme are clearly superior to those of the unweighted linear model. Not only is the noise in the

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shadow areas diminished, but the overall picture is crisper and displays more contrast, with the fine detail much more evident. This result indicates that, for no extra expenditure in transmission rate, the image quality can be markedly improved by simply preprocessing with the best model before transmission and postprocessing with its inverse after reception. This "pre-emphasis" technique in fact is a time-honored method used by engineers for improving the quality of signals in such diverse areas as FM transmission, magnetic tape recording, and radar signal processing.

There are several extensions to this modeling technique which might be profitably explored to increase the usefulness and accuracy of the model. First, a more complex form for the model might be used, more closely representing the various optical and neural transfer functions. Secondly, the phenomenon of independent frequency channels in the visual system might be examined and an alternative to integral-weighted-square error considered. Finally, and probably of greatest importance, the interaction between the image and the error and the manner in which the visual system actually scans an image should be taken into account.

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